NEQRX: A Quantum Image Encryption and Reversible Data Hiding Scheme With Reduced Circuit Complexity

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***Abstract*—The development of image processing requires advanced cryptographic and embedding techniques to ensure data security and efficiency. However, existing approaches suffer from high quantum circuit complexity, limited scalability, and increased vulnerability to both classical and quantum threats. Therefore, we introduce NEQRX, an efficient quantum image encryption and reversible data hiding (RDH) scheme that combines the Generalized Affine Transform (GAT) and Logistic Map (LM) to enhance security and data embedding capacity while significantly reducing quantum circuit complexity and cost. We introduce a novel two-stage optimization framework that first applies the classical Espresso algorithm and then uses the ancillary qubit compression technique. This hybrid approach achieves cumulative reductions exceeding 60% in quantum cost and 64% in time complexity, significantly outperforming optimization from the Espresso algorithm alone while maintaining both security and efficiency. Comprehensive experimental evaluations on Qiskit's quantum simulators validate the practical realization of our approach. NEQRX outperforms other state-of-the-art techniques in data embedding capacity and image recovery fidelity. Furthermore, our RDH algorithm maintains high image quality and embedding capacity, outperforming other approaches in terms of peak signal-to-noise ratio (PSNR) as the embedding rate increases. Finally, the security and robustness analysis of NEQRX demonstrates its resilience against various noise models, ensuring a scalable and efficient framework that enhances the security of digital image processing.**

***Index Terms*—Novel enhanced quantum representation (NEQR), generalized affine transform (GAT), logistic map (LM), quantum cost, complexity analysis, noisy channels, and fidelity.**

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I. Introduction

DIGITAL image processing has become a basis of information science, playing a pivotal role in diverse fields such as remote sensing, robotic vision, pattern recognition, and medical imaging [1]. At its core, digital image processing involves the analysis, transmission, and manipulation of digital imagery, represented as a two-dimensional array (matrix) of numbers obtained by sampling and quantizing analog images. The exponential growth in digital image acquisition and sharing, combined with the widespread implementation of machine learning, the Internet of Things, and other technologies, has increased the demand for the quality and security of digital images. The quality of digital images is determined by factors such as resolution and pixel count, while security focuses on preventing unauthorized access to the information they contain. For security, the development of strong cryptographic techniques can withstand unauthorized access. Reversible data hiding (RDH) has emerged as a key method to enhance image security and integrity. With RDH, secret data can be hidden in images, and the original image is fully restored after data extraction [2]. This functionality is particularly advantageous in applications such as legal forensics, military images, and medical imaging, where both the hidden data and the original image are paramount. Traditional RDH methods in classical computing have been studied by researchers for some time. These methodologies have essentially been grouped into three categories: lossless compression-based algorithms [3], difference expansion-based algorithms [4], and histogram shifting-based algorithms [5]. With quantum computing becoming established, the need for RDH methods that can withstand quantum attacks or operate on quantum image representations is increasing.

Classical cryptosystems, such as the Data Encryption Standard (DES), Triple-DES, Advanced Encryption Standard (AES), Elliptic Curve Cryptography (ECC), and Rivest-Shamir-Adleman (RSA), have been widely used for secure encryption. Although these algorithms are practical for traditional encryption, they are increasingly vulnerable to the emerging threat of quantum computing [6]. Quantum computers have already demonstrated the ability to perform real-time factorization using quantum algorithms, thereby undermining the security of classical cryptographic methods. Moreover, as quantum computers continue to prove their superiority over classical methods, they will likely become a reality. Therefore, to ensure the confidentiality and integrity of digital imagery in the quantum computing era, it is necessary to develop encryption and data hiding techniques based on quantum computing principles and achieve post-quantum cryptography [7]. Several methods have been proposed for computing and storing digital images on quantum computers [8]. These include techniques such as qubit lattice representation [9], real ket representation [10], entangled image encoding [11], flexible representation of quantum images (FRQI) [12], multi-channel quantum image representation (MCQI) [13], novel enhanced quantum representation (NEQR) [14], novel quantum representation of color digital images (NCQI) [15], a generalized NEQR model (GNEQR) [16], normal arbitrary quantum superposition state (NAQSS) [17], and quantum Boolean image processing (QuBoIP) [18]. The NEQR method is similar to FRQI but encodes pixel values differently. NEQR uses eight qubits to encode 256 different pixel values, while FRQI relies on the polar and azimuthal angles of a single-qubit state. This difference allows NEQR to achieve a higher measurement accuracy than FRQI.

Advances in quantum image encryption algorithms have strengthened the protection of image data through quantum operations. These algorithms encrypt the image data to produce a cipher image derived from the original image by scrambling pixel positions and changing pixel values, often using chaos theory or other complex transformations. These encryptions fall into two categories: 1) transformation in the frequency domain with random operations [19], [20], [21] and 2) encryption of the image directly using chaos theory [19], [22], [23]. This work concerns the second method, in which pixel values are encrypted using a logistic map (LM), and pixel positions are scrambled using a generalized affine transformation (GAT), with an additional scheme for quantum-reversible data hiding. Traditional image scrambling techniques, such as Hilbert [24], magic cube [25], and Arnold and Fibonacci transformations [26], are widely used in classical computing for applications such as secure distribution of digital image transmission, cryptographic applications for safe storage, and watermarking [26], [27], [28]. The central contributions of chaos theory, previously applied in image processing and the development of quantum encryption, have expanded and redefined traditional image processing cryptography, which now involves more complex scrambling techniques and designs to increase security as advances in quantum computing progress. For example, XOR operations based on chaotic systems have been used to scramble color information in images, and bit-by-bit operations combined with key images generated from chaotic sequences have enhanced encryption effectiveness by increasing key space and reducing computational complexity.

In this paper, we propose an efficient implementation scheme for a quantum encryption technique. This technique uses NEQR for encoding [14], and combines GAT to scramble pixel positions [19] and LM based on chaos theory to alter pixel values [19] for encryption. Additionally, a reversible data hiding scheme is used for secure data embedding. To optimize circuit complexity, we employ the Espresso algorithm [29], as used in [14], and present an ancillary qubit technique. Finally, we evaluate the encryption and data embedding scheme by performing data embedding and statistical analysis, calculating PSNR and SSIM, and analyzing circuit complexity and fidelity variation in the NEQR circuit across different levels of quantum and classical noise.

The contributions of this article can be summarized as follows:

1) We propose an efficient scheme for quantum image encryption and reversible data hiding based on GAT and LM, enhancing security and data embedding capacity.
2) We introduce a novel hybrid optimization framework that combines the classical Espresso algorithm and the ancillary qubit technique. This framework achieves cumulative reductions exceeding 60% in quantum cost and 64% in time complexity, demonstrating a significant optimization that surpasses what the Espresso algorithm alone can achieve.
3) We provide a comprehensive experimental evaluation, comparing our method with other state-of-the-art techniques, and demonstrate superior performance in terms of data embedding capacity and image recovery fidelity.

The remainder of this paper is organized as follows: Section II provides a brief review of the preliminaries. Section III describes the proposed quantum image encryption algorithm, and Section IV introduces the data embedding and extraction method. Section V presents the evaluation of the encryption and data embedding scheme. Finally, Section VI concludes the paper.

II. Preliminaries

A. Novel Enhanced Quantum Representation (NEQR)

NEQR, introduced by Zhang et al. [14], offers a novel approach to quantum gray image representation. We use this approach to represent quantum gray images using the gray image parameters as input and to encode them into a quantum circuit (see Algorithm (1) in the supplementary material and Fig. 1). The construction of this circuit involves $2n + 8$ qubits, with 8 qubits designated to store the grayscale value $f(Y, X)$ of the pixels, while the remaining $2n$ qubits represent the pixel positions (Y, X). The resulting state of this quantum circuit encapsulates the entire image. Mathematically, the final state can be expressed as follows:

$$|\psi\rangle = \frac{1}{2^n} \sum_{Y,X=0}^{2^{2n}-1} |f(Y,X)\rangle\, |YX\rangle\,, \tag{1}$$

where $|f(Y,X)\rangle$ can be written in the bit sequence as $\bigotimes_{i=0}^{p-1} |C_{YX}^{i}\rangle = |c_{YX}^{7} c_{YX}^{6} \ldots c_{YX}^{1} c_{YX}^{0}\rangle$, which represents the encoded gray-scale value of the pixel, and

$$\begin{aligned} |YX\rangle &= |Y\rangle|X\rangle \\ &= |Y_{n-1}Y_{n-2}\ldots Y_1Y_0\rangle|X_{n-1}X_{n-2}\ldots X_1X_0\rangle, \end{aligned}$$

is used to store the corresponding vertical and horizontal coordinate values of the pixels.

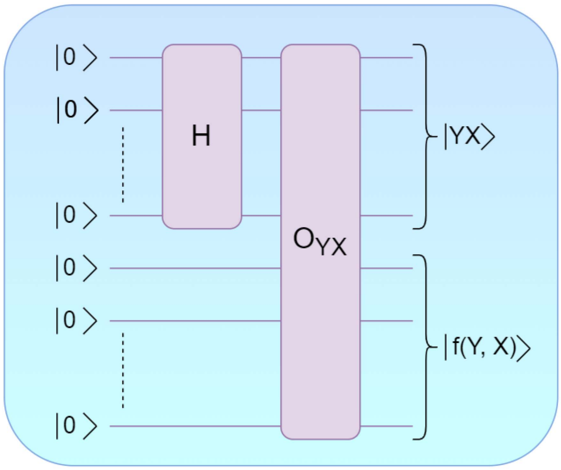


Fig. 1. NEQR procedure circuit using $(2n+8)$-qubits. Here, H is the Hadamard gate and O_{YX} is a controlled gate that operate U_{YX} on pixel qubits, when 2n-qubits are in the state $|YX\rangle$, to obtain $f(Y,X)$.

For example, upon performing NEQR on an image of size 2×2, it can be represented by the quantum state:

$$|\psi_I\rangle = \frac{1}{2}(|11111111\rangle\,|00\rangle + |00000000\rangle\,|01\rangle + |11001000\rangle\,|10\rangle + |01100100\rangle\,|11\rangle). \quad (2)$$

B. Generalized Affine Transform

The GAT is used to scramble the image's pixel locations $|YX\rangle$ (see Algorithm (2) in the supplementary material). Here, X and Y represent the actual horizontal and vertical locations of pixels, respectively [19]. The mathematical representation of GAT is given by the following:

$$\begin{pmatrix} X' \\ Y' \end{pmatrix} = \begin{pmatrix} s & 0 \\ 0 & t \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} + \begin{pmatrix} P \\ Q \end{pmatrix} \mod 2^n, \quad (3)$$

where X' and Y' represent the scrambled horizontal and vertical locations of pixels, taking the forms $(sX+P)$ and $(tY+Q)$, respectively, with modulo 2^n. The parameters s, t, P & Q are for GAT and obey the following conditions:

- P and Q should not be zero. In the circuit implementation, n qubits are allocated to each qubit to store its value.
- s and t should be co-prime with 2^n.

The generalized circuit of the operations involved in scrambling the original coordinates $|X\rangle, |Y\rangle$ to $|X'\rangle, |Y'\rangle$ is illustrated in Fig. 2. The inverse operation of GAT is utilized to recover the original pixel coordinates from the scrambled ones (see Algorithm 3 in the supplementary material). Mathematically, this inverse operation can be expressed as:

$$\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} s^{-1} & 0 \\ 0 & t^{-1} \end{pmatrix} \begin{pmatrix} X'-P \\ Y'-Q \end{pmatrix} \mod 2^n. \quad (4)$$

Equation (4) gives the original coordinates of the pixel $X = s^{-1}(X'-P)\,mod\,2^n$ and $Y = t^{-1}(Y'-Q)\,mod\,2^n$. Here, s^{-1} and t^{-1} are the modular multiplicative inverse [30] of s and t, respectively, and can be calculated as follows:

$$1 = ss^{-1} \mod 2^n,$$
$$1 = tt^{-1} \mod 2^n. \quad (5)$$

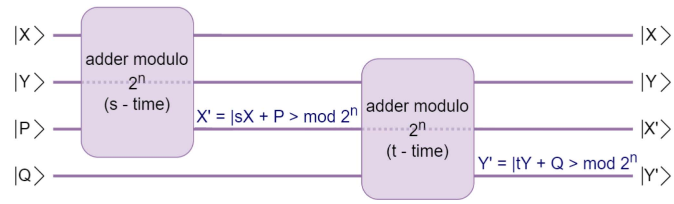


Fig. 2. A quantum circuit for GAT to scramble the original pixel coordinate $|YX\rangle$ into $|Y'X'\rangle$. The dotted qubit line in the circuit represents bypassing the operation.

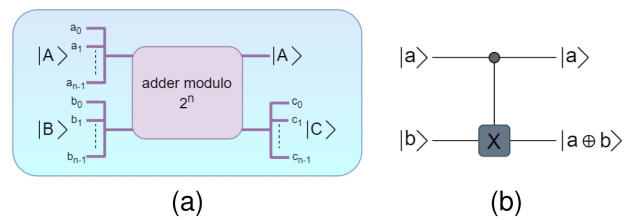


Fig. 3. (a) Adder modulo circuit: with output $|C\rangle = |A\rangle \oplus |B\rangle \mod 2^n$. (b) Quantum CNOT gate: gives output as $a \oplus b\; mod\, 2$.

The generalized circuit in Fig. 4 shows the operations performed to recover the original coordinates $|X\rangle, |Y\rangle$ from $|X'\rangle$, $|Y'\rangle$ as calculated in (4).

C. Adder Modulo 2^n

The adder modulo 2^n is designed to sum two binary numbers, each represented by n bits, ensuring that the resultant binary number maintains the same number of bits. This is achieved by ignoring the last carry bit. A generalized quantum circuit for the adder modulo 2^n is given in [20]. A general black-box representation of this circuit for $2n$ bits input is shown in Fig. 3(a). For $n = 1$, the adder modulo 2^n circuit simplifies to the quantum CNOT gate, shown in Fig. 3(b).

D. Logistic Map

LM is used to iteratively map the value of the previous step to the next step [22], using (6):

$$L_{\eta+1} = \delta L_\eta (1 - L_\eta), \quad (6)$$

where $\eta = 0, 1, 2, \ldots, 2^{2n} - 1$. Here, $L_0 \epsilon 0, 1$ denotes the initial value of LM, δ represents the growth rate, and 2^{2n} signifies the total number of pixels. This dynamic equation exhibits chaotic behavior for $3.85 \le \delta \le 4$, which means that with this growth rate, (6) generates pseudo-random strings of 0's and 1's.

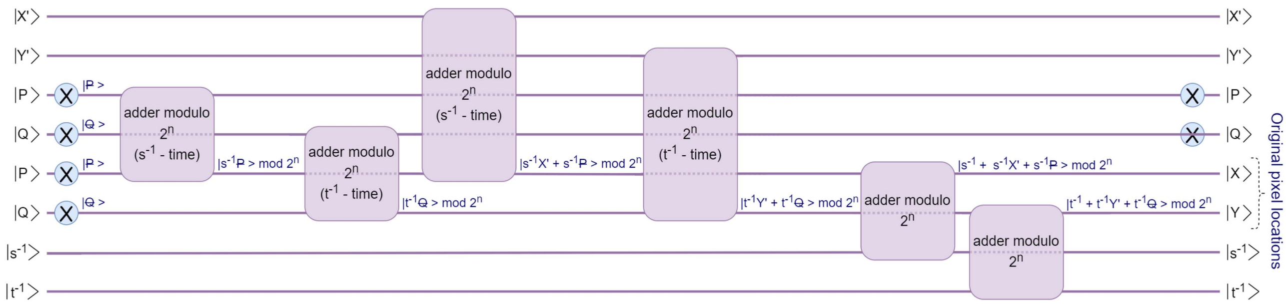


Fig. 4. Quantum circuit represents a general decryption procedure for GAT to retrieve the real pixel coordinate $|YX\rangle$ from $|Y'X'\rangle$.

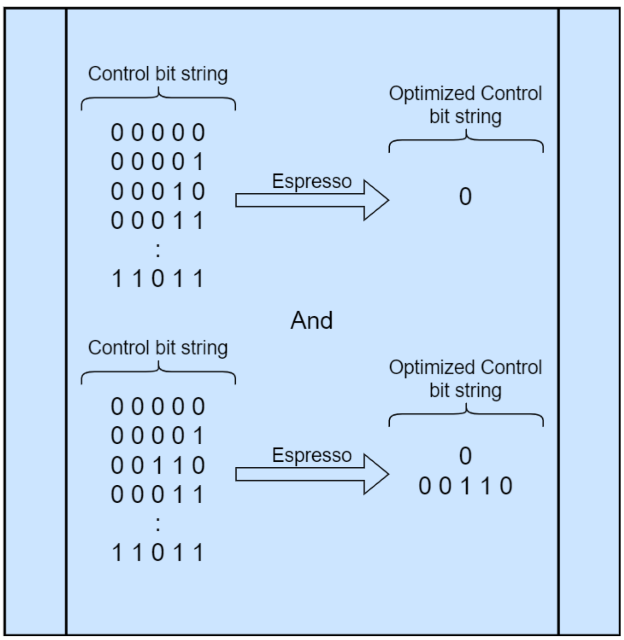


Fig. 5. An example to demonstrate the optimization of control bit strings using the Espresso algorithm. The original control bit strings on the left and the optimized control bit strings on the right can perform the same operation on the i^{th} target qubit.

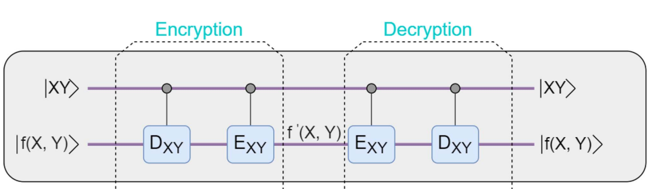


Fig. 6. LM for encryption and decryption of pixel value guided by horizontal X and vertical Y position of pixels.

III. Proposed Algorithm

A. NEQR Circuit

A quantum circuit where $n = 1$ of $2 + 8$ -qubits is initialized in the all-0s state, as shown in Fig. 9(a). The first 2 qubits store the pixel coordinates, while the remaining qubits represent the pixel values. The encoding procedure of the image parameters in a quantum circuit is as follows:

1) The 2×2 image has 4-pixels with different positions and values. To process all pixel positions, the Hadamard gates on the first 2 qubits are used to generate a superposition of four equally probable states ($|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$).
2) Toffoli gates are then applied, using coordinate qubits as the control and the remaining qubits as targets. These Toffoli gates transform the pixel value qubits from state $\bigotimes_{i=0}^{7} |0^i\rangle$ to state $\bigotimes_{i=0}^{7} |C_{YX}^i\rangle$, corresponding to each pixel $|YX\rangle$.

The conversion of the state of the quantum circuit is as follows:

$$|0000000000\rangle \xrightarrow[\text{first 2-qubits}]{\text{apply H-gate on}} |00000000\rangle\,|{+}{+}\rangle \xrightarrow[\text{on rest qubits}]{\text{apply Toffoli gates}} |\psi_I\rangle\,. \tag{7}$$

Fig. 9(a) shows the resultant quantum circuit of this procedure, featuring 14 Toffoli gates. However, before executing on real superconducting devices, the circuit must be transpiled to the device's basis gates, which can exponentially increase circuit size and depth, impacting execution efficiency and fidelity. To solve this, we employ a two-stage optimization framework. In the first stage, we apply the image compression technique from [14], which utilizes the well-established Espresso algorithm [29]. It processes the control bit strings of pixel values to perform the same task with reduced control bit strings, as shown in Fig. 5. As a result, optimization minimizes circuit complexity. In the second stage, we employ an ancillary qubit compression technique to further reduce circuit complexity. This technique utilizes an ancillary qubit to store information from multiple Toffoli gates with identical control bit strings, which are then systematically applied as control and target qubits using CNOT gates. The same task can be implemented using fewer Toffoli gates, CNOT gates, and a single ancillary qubit (see Fig. 9(c)). This second stage is particularly effective when the number of Toffoli gates with identical control bit strings remains high after applying the Espresso algorithm.

B. Encryption Procedure

The encryption procedure utilizes a combination of LM and GAT to transform the original image into an encrypted form. The choice of LM and GAT is directly aligned with the two-register structure of NEQR. As shown in (1), an NEQR image is encoded as $|f(Y,X)\rangle|YX\rangle$, which separates pixel intensity from spatial

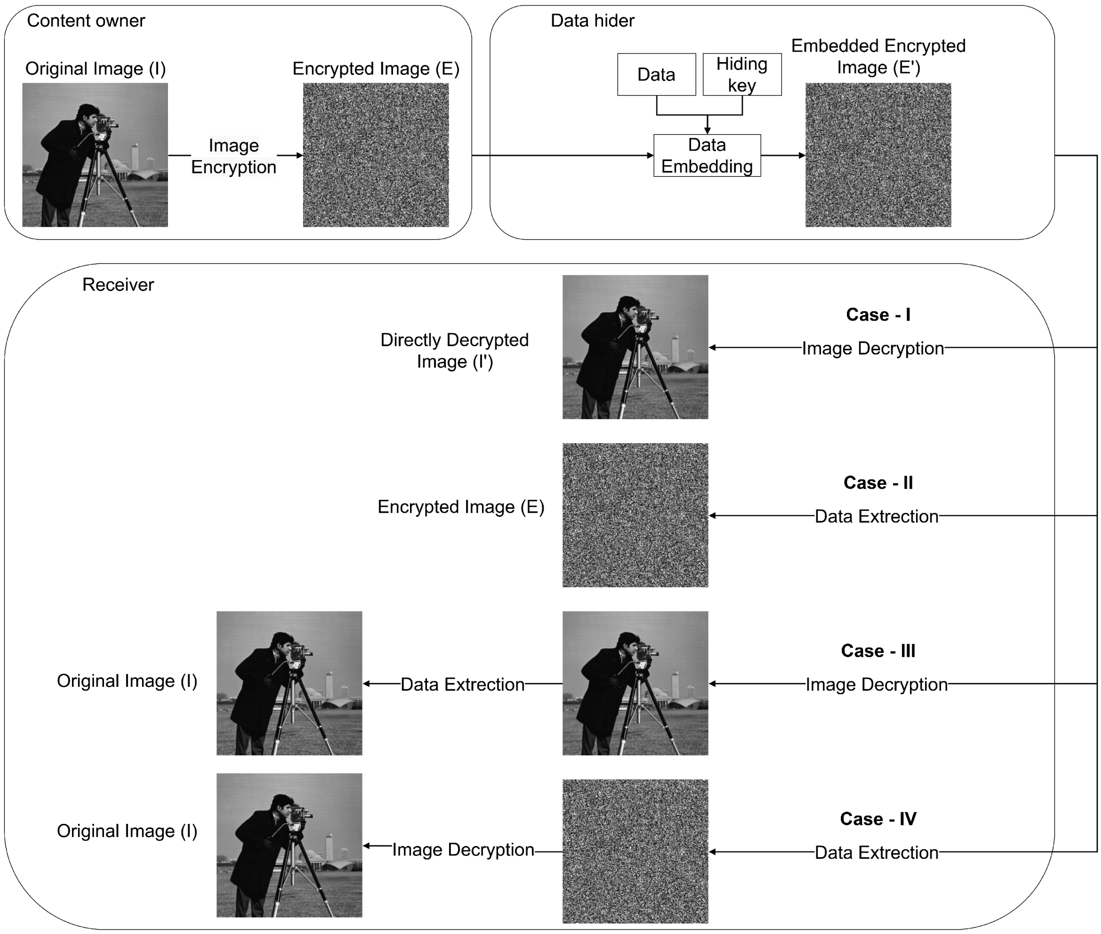


Fig. 7. The overall framework of the proposed scheme includes image encryption, data hiding, and the four cases for data extraction and image recovery. The experimental transformation of the original 512 × 512 'Cameraman' image and its histogram for this scheme are presented in Fig. 10.

coordinates. The LM-based diffusion stage acts on the gray-value register through reversible XOR/CNOT-type operations controlled by the corresponding coordinate state (Section III-B1) whereas the GAT permutation stage acts on the coordinate register through reversible modular arithmetic (3)–(5). Both operations are unitary and invertible, preserving reversibility for exact decryption and image recovery. The transformations are therefore not arbitrary classical mappings forced onto a quantum state; each acts naturally on the register that carries the information it is designed to scramble.

1) Logistic Map (Diffusion Stage): The diffusion stage is applied to the pixel value and is controlled by the corresponding pixel coordinates. This stage employs chaos theory, producing a chaotic image. The quantities required to operate LM in a quantum circuit are calculated according to the Algorithm 1. The operations of LM in a quantum circuit are as follows:

1) Pick the output lists J, T of Algorithm 1.
2) Transform each J_{YX} and T_{YX} form J and T to a binary sequence as:

$$|J_{YX}\rangle = \otimes_{i=0}^{p-1} \left|J_{YX}^i\right\rangle, \quad |T_{YX}\rangle = \otimes_{i=0}^{p-1} \left|T_{YX}^i\right\rangle. \quad (8)$$

3) Defining quantum operations D_{YX} and E_{YX} to manipulate pixel values based on J_{YX} and T_{YX} as:

$$D_{YX}\left|C_{YX}^i\right\rangle \to \left|C_{YX}^i \oplus J_{YX}^i\right\rangle,$$
$$E_{YX}\left|C_{YX}^i \oplus J_{YX}^i\right\rangle \to \left|C_{YX}^i \oplus J_{YX}^i \oplus T_{YX}^i\right\rangle.$$

4) Define the functions that apply D_{YX} and E_{YX},

$$\phi_{YX}^1 = I \otimes \frac{1}{2^n} \sum_{j=0}^{2^n-1} \sum_{i=0, ji \neq YX}^{2^n-1} |ji\rangle\langle ji| + D_{YX} \otimes |YX\rangle\langle YX|,$$

$$\phi_{YX}^2 = I \otimes \frac{1}{2^n} \sum_{j=0}^{2^n-1} \sum_{i=0, ji \neq YX}^{2^n-1} |ji\rangle\langle ji| + E_{YX} \otimes |YX\rangle\langle YX|.$$

5) Apply ϕ_{YX}^1 and ϕ_{YX}^2 to the quantum state ψ_I as:

$$\begin{aligned} |\psi_L\rangle &= \phi_{YX}^2 \phi_{YX}^1 |\psi_I\rangle \\ &= \frac{1}{2^n} \sum_{Y,X=0}^{2^{2n}-1} \left(E_{YX} D_{YX} \bigotimes_{i=0}^{p-1} \left|C_{YX}^i\right\rangle\right) |YX\rangle \end{aligned}$$

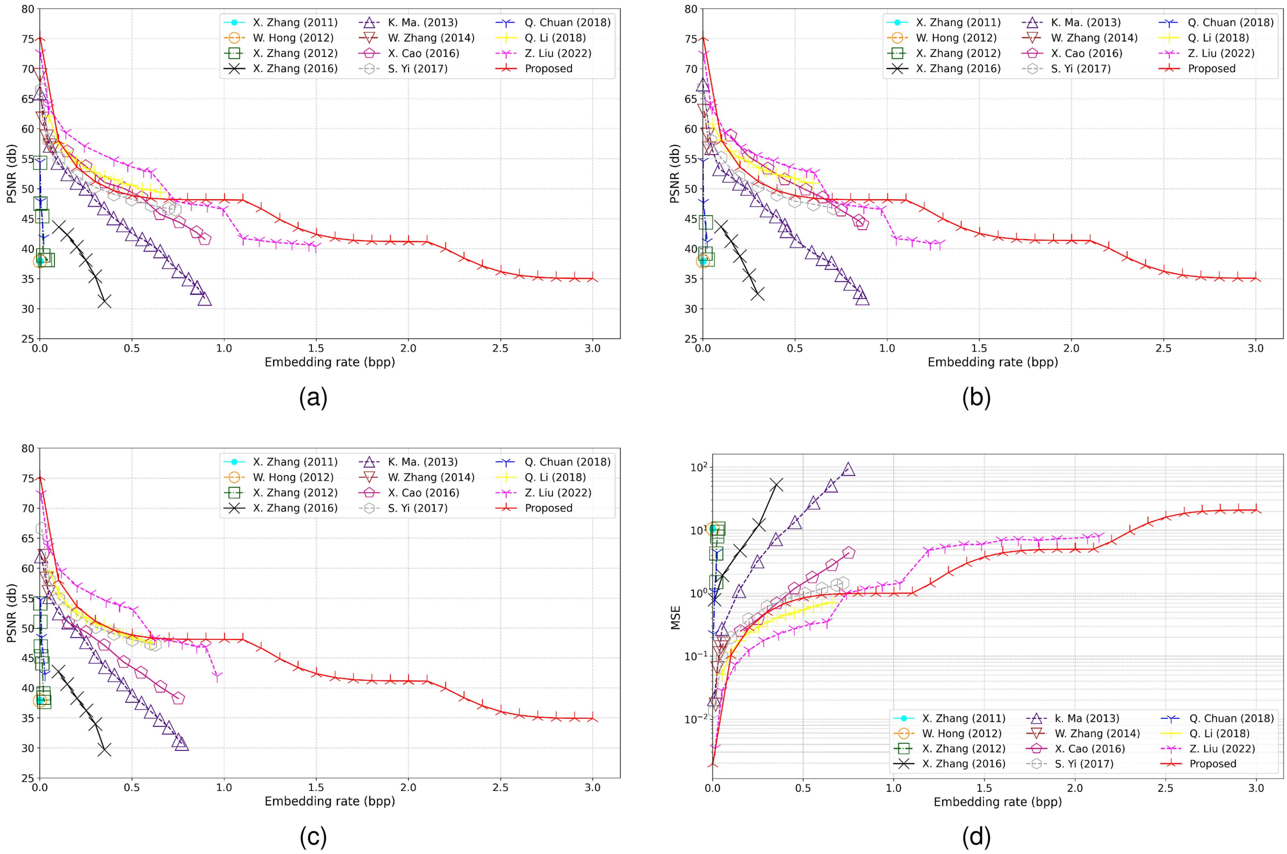


Fig. 8. The variation in PSNR for directly decrypted images is compared to data from previous research. Subfigures (a), (b), and (c) illustrate the PSNR as a function of bits per pixel (bpp) for the Airplane, Crowd, and Man images, respectively. Subfigure (d) presents the average mean squared error (MSE) versus bpp, derived from the results in (a), (b), and (c).

$$= \frac{1}{2^n} \sum_{Y,X=0}^{2^{2n}-1} \bigotimes_{i=0}^{p-1} \left| C^i_{YX} \oplus J^i_{YX} \oplus T^i_{YX} \right\rangle |YX\rangle .$$

2) Generalized Affine Transform (Permutation Stage): GAT, represented by the operator A, operates only on the pixel positions of the quantum state of the image $|\psi_L\rangle$. The operation is as follows:

$$\begin{aligned} |\psi_{AL}\rangle = A\,|\psi_L\rangle &= \frac{1}{2^n} \sum_{Y=0}^{2^n-1} \sum_{X=0}^{2^n-1} |f'(Y,X)\rangle\, A\,|YX\rangle \\ &= \frac{1}{2^n} \sum_{Y=0}^{2^n-1} \sum_{X=0}^{2^n-1} |f'(Y,X)\rangle\, |Y'X'\rangle , \end{aligned}$$

where A scrambles the horizontal and vertical positions of the pixels as $A|X\rangle = |X'\rangle$ and $A|Y\rangle = |Y'\rangle$, respectively. The output of this algorithm $|X'\rangle$ and $|Y'\rangle$ is given by:

$$\begin{aligned} |X'\rangle &= |sX + P\rangle \mod 2^n, \\ |Y'\rangle &= |tY + Q\rangle \mod 2^n. \end{aligned}$$

According to the quantum version of GAT, which scrambles the pixel positions, a general quantum circuit is designed in Fig. 2. This quantum circuit performs the adder modulo 2^n operation, s and t times to transform the states $|X\rangle \rightarrow |X'\rangle$ and $|Y\rangle \rightarrow |Y'\rangle$, respectively, as shown in Fig. 2. The algebra of a quantum circuit is as follows:

$$\begin{aligned} |X, P\rangle &\xrightarrow[\text{s - times}]{\text{adder modulo } 2^n} |X, (sX + P) \mod 2^n\rangle , \\ |Y, Q\rangle &\xrightarrow[\text{t - times}]{\text{adder modulo} 2^n} |Y, (tY + Q) \mod 2^n\rangle . \end{aligned}$$

C. Decryption Procedure

The decryption process recovers the original image by sequentially applying the inverse transformation of the operations used in encryption.

1) Inverse Generalized Affine Transform: The inverse GAT retrieves the original horizontal and vertical pixel coordinates of the image. Let A^{-1} be the inverse GAT operation:

$$\begin{aligned} |\psi_L\rangle &= A^{-1}\,|\psi_{AL}\rangle \\ &= \frac{1}{2^n} \sum_{Y,X=0}^{2^{2n}-1} |f'(Y,X)\rangle\, A^{-1}\,|Y'X'\rangle \\ &= \frac{1}{2^n} \sum_{Y,X=0}^{2^{2n}-1} |f'(Y,X)\rangle\, |YX\rangle . \end{aligned}$$

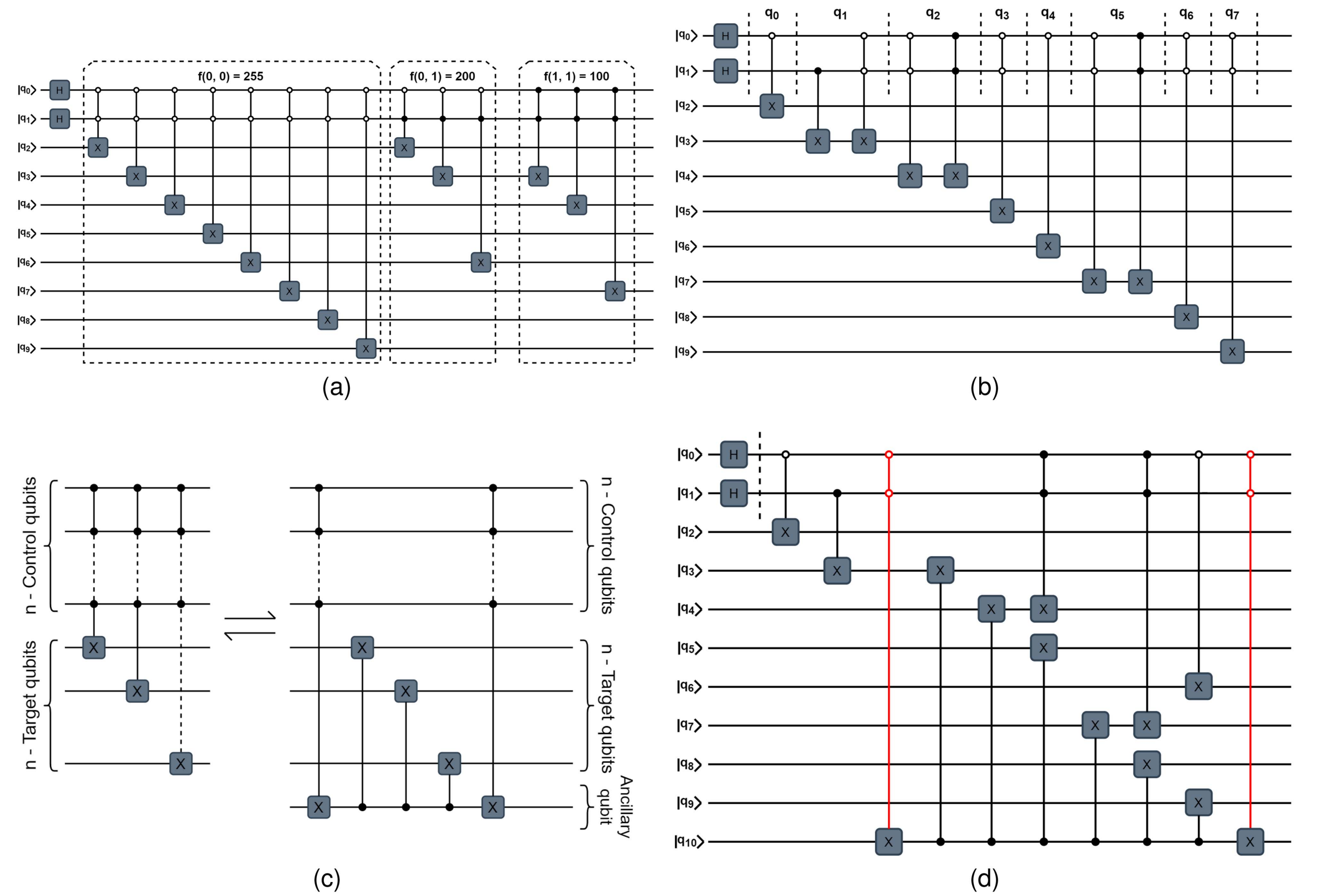


Fig. 9. (a) The baseline quantum circuit for the image represented by (2), according to the NEQR quantum model. Note that the position coordinates of the pixel are encoded into the circuit as $|YX\rangle \rightarrow |q_1 q_0\rangle$ and the pixel values as $|C^0_{YX} C^1_{YX} \dots C^7_{YX}\rangle \rightarrow |q^9 q^8 \dots q^2\rangle$. (b) The circuit in Fig. 9(a) after the reduction in the controlled information for the pixel values using the Espresso algorithm. (c) Conceptual diagram of the ancillary qubit technique, which reduces the cost of multiple Toffoli gates sharing an identical control bit string. (d) The final, fully optimized circuit after applying our complete hybrid framework (Espresso followed by the ancillary qubit technique). The Toffoli gate, represented by red, is the one whose information is saved in the ancillary qubit.

Algorithm 1: Encrypting Pixel Values using LM.

Input: Declare the size of the image: $2^n \times 2^n$
  Declare the LM parameters: δ, L_0
Output: J_{YX}, T_{YX} for all pixel positions $YX = \eta$
  Initialisation:
    $J, T \leftarrow [\ \ ]$
  GENERATE (J_{YX})
    For all η **in range** $(2^{2n} - 1)$ **do**
      $L_{\eta+1} \leftarrow \delta \cdot L_\eta \cdot (1 - L_\eta)$
      $J_{YX} \leftarrow \text{round}(\text{mod}(L_\eta \times 2^8, 2^8))$
      $J.\text{append}(J_{YX})$
  GENERATE (T_{YX})
    For all η **in range** $(2^{2n} - 1)$ **do**
      $T_{YX} \leftarrow J_{2^{2n}-1-YX}$
      $T.\text{append}(T_{YX})$
    return J, T

According to Theorem 3 in [20], the negative adder modulo 2^n in (4) can be realized as:

$$(x - y) \mod 2^n = (x + (\overline{y} + 1)) \mod 2^n,$$

where $\overline{y} = \overline{y}_{n-1}\overline{y}_{n-2}\dots\overline{y}_0 = 1$ - y_i with $i = n-1, n-2, \dots, 0$. Thus, $X = s^{-1}(X' - P)\ mod\ 2^n$ is converted to $X = s^{-1}(X' + (\overline{P} + 1))$ and similarly for Y. The circuit in Fig. 4 takes 2 $s^{-1} + 3$ steps to obtain the original pixel position $|X\rangle$ and $2t^{-1} + 3$ for $|Y\rangle$.

2) Logistic Map: The inverse LM procedure is straightforward, with the output provided by Algorithm 1. This procedure operates the same functions ϕ^1_{YX} and ϕ^2_{YX} used in the encryption procedure (see Fig. 6) as:

$$\begin{aligned} |\psi\rangle &= \phi^1_{YX}\phi^2_{YX}\,|\psi_L\rangle \\ &= \phi^1_{YX}\phi^2_{YX}\phi^2_{YX}\phi^1_{YX}\,|\psi\rangle \\ &= \frac{1}{2^n}\sum_{Y,X=0}^{2^{2n}-1}\left(D_{YX}E_{YX}E_{YX}D_{YX}\bigotimes_{i=0}^{p-1}\left|C^i_{YX}\right\rangle\right)|YX\rangle \\ &= \frac{1}{2^n}\sum_{Y,X=0}^{2^{2n}-1}\bigotimes_{i=0}^{p-1}\left|C^i_{YX}\right\rangle|YX\rangle. \end{aligned} \tag{9}$$

IV. Data Embedding

Once the data hider acquires the encrypted image $|\psi_{AL}\rangle$, they can embed the desired data into it, although they lack access to the original image $|\psi\rangle$. The embedding process begins by generating an all-zero matrix of the same size as the encrypted image. This matrix will serve as a data storage medium. The data hider proceeds by inserting each bit of the desired data into the elements of this matrix, starting from the first element and moving sequentially through the matrix. Once the data has been embedded in the matrix, the data hider will apply an affine transformation to randomize the matrix elements, thereby protecting the embedded data. After randomization is completed, the data hider will begin quantum embedding. The randomized matrix with embedded data will serve as the control bits, and the LSBs (least significant bits) of the encrypted image $|\psi_{AL}\rangle$ will serve as the target bits; these two bits are what secure hidden data in $|\psi_{AL}\rangle$. After this process, the data hider is left with an encrypted, data-embedded image and the data-hiding key. This key will include the scrambling pattern and the order in which to apply the CNOT, both of which are necessary for the data extraction process and for the correct retrieval of the embedded data. The mathematical explanation is as follows:

1) *Initialize Zero Matrix:* Define a zero matrix M with the same dimensions as the encrypted image $|\psi_{AL}\rangle$

$$M = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}$$

2) *Embed Desired Data into Matrix M:* Insert the desired data d_i sequentially into the elements of M:

$$M(i,j) = d_{ij} \quad \text{where } d_{ij} \in \{0, 255\}$$

where i, j are the indices that iterate through the matrix.

3) *Affine Transformation:* To secure the embedded data, apply an affine transformation on M:

$$M' = a \cdot M + b \mod N$$

where a and b are constants that define the affine transformation, and M' is the scrambled version of M. N represents the modular base.

4) *Quantum CNOT Embedding:* After scrambling, use the matrix M' as control bits and the least significant bits (LSBs) of the encrypted image $|\psi_{AL}\rangle$ as target bits in a sequence of CNOT operations:

$$\text{CNOT}(|\psi_{AL}\rangle, M') = |\psi_{AL}^{(E)}\rangle = E'$$

where $|\psi_{AL}^{(E)}\rangle$ represents the encrypted image embedded in data. Each element $M'(i,j)$ in the scrambled matrix serves as the control bit for the corresponding encrypted pixel in $|\psi_{AL}\rangle$.

5) *Data Hiding Key:* The data hiding key, K, consists of the affine transformation parameters (a, b) and the sequence of CNOT operations. This key is essential for retrieving the embedded data:

$$K = \{a, b, \text{CNOT sequence}\}$$

A. Data Extraction and Image Recovery

In the extraction phase, data extraction and image recovery are independent operations, allowing different levels of access based on available keys. After the encrypted image embedded in the data $|\psi_{AL}^{E}\rangle$ is passed to a receiver, three cases are possible:

1) *The Receiver Has Only the Encryption Key (case-I):* In this case, the receiver proceeds as described in Section III-C). Using the encryption key to decrypt each pixel value, the receiver can obtain a directly decrypted image, I', which maintains a high peak signal-to-noise ratio (PSNR) relative to the original image I. It is essential to note that the hidden data remain embedded in I' and are not affected by this decryption process.

2) *The Receiver Has Only the Data Hiding Key (case-II):* If the receiver only has access to the data hiding key, they can extract the embedded data but will not be able to decrypt the image. The receiver first reconstructs the control matrix from the data that hides the key's CNOT sequence. Using this matrix, the receiver performs a quantum CNOT operation on $|\psi_{AL}^{E}\rangle$, with the matrix elements as control bits and the least significant bits (LSBs) of each pixel in $|\psi_{AL}^{E}\rangle$ as target bits. The receiver then compares the new LSB values with the original LSBs in $|\psi_{AL}^{E}\rangle$, constructing a difference matrix by recording the differences in each pixel's LSB. Specifically, if both bits are the same (either both 0 or both 1), the difference is 0; if they differ, the difference is 1. Finally, the receiver applies the inverse affine transformation (using the affine transform parameters from the data hiding key) to decode and retrieve the original data from this difference matrix.

3) *The Receiver Has Both the Data Hiding Key and the Encryption Key:* In this case, the receiver has full access to both the original data and the original image. The receiver can choose to extract the data first, as described in case-II, or decrypt the image first, as described in Section III-C and case-I. Following this initial step, the remaining operation will yield the second piece of information. Consequently, the receiver will finally obtain both the original data and the original image. This dual approach is represented in Fig. 7 as case-III for decryption followed by data extraction and case-IV for data extraction followed by decryption.

V. Evaluation

Image reconstruction is performed using the pixel values and pixel positions obtained after simulation of quantum circuits (constructed according to Fig. 1) in the Qiskit Qasm simulator, which resembles a real perfect quantum computer [31].

A. Data Embedding Analysis

To evaluate the effectiveness of our proposed RDH algorithm, we conducted a comprehensive comparison with other state-of-the-art RDH algorithms [2], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41]. For the sake of fairness, these statistical results are directly derived from the original data reported in the literature or obtained by further averaging of the reported results. This comparison focuses on two key metrics: the Peak Signal-to-Noise Ratio (PSNR), which represents the quality of the image after data embedding compared to the original, and embedding capacity, measured in bits per pixel (bpp), indicating the number of bits that can be embedded per pixel. Our experiments utilized three standard images of size 512×512: Airplane, Crowd, and Man.

Here, we used an embedding dataset consisting entirely of ones, ensuring that the Controlled-NOT (CNOT) operation applied to each pixel activates the gate for every bit, resulting in the maximum possible modification of the least significant bits (LSB). This setup allowed us to observe the maximum change and rigorously test our algorithm's embedding capacity. We achieve lower PSNR values than other algorithms at lower embedding rates, as shown in Fig. 8(a)–(c). However, as the embedding rate increases, our PSNR remains stable and exceeds that of all other algorithms. We observed a PSNR of 35 dB at an embedding rate of 3 bpp, significantly higher than those of other methods, even at embedding rates below 1 bpp. This demonstrates our algorithm's superior ability to maintain image quality and a higher embedding capacity.

Furthermore, we calculate the average Mean Squared Error (MSE) for the three images, as shown in Fig. 8(d). The results demonstrate that our proposed RDH algorithm outperforms existing state-of-the-art methods in both embedding capacity and image quality. This makes it highly suitable for applications that require secure, high-capacity data embedding without compromising the integrity of the original image.

1) Reversibility: According to the reversibility principle, if an authorized user can perfectly reconstruct the original image from the marked image, the method used to mask the image is reversible. To verify this, we employ two widely used metrics: PSNR and SSIM.

Table II presents the quantitative results for several test images. The comparison between the original and recovered images yields a PSNR of infinity ($+\infty$) and an SSIM of 1 in all images tested. These results confirm the reversibility of the proposed method, demonstrating complete preservation of original image information.

B. Performance Analysis

1) Embedding Capacity Analysis: The embedding capacity refers to the number of secret data bits that can be hidden in an encrypted image. It is measured by the embedding rate (ER) in bits per pixel (bpp), calculated as ER = Total hidden bits / Total pixels in the cover image. In our method, CNOT-based data fusion allows hiding secret data of the same size as the original image within the encrypted image. This gives a maximum embedding capacity equal to the cover image size. For example, we used a 512×512 grayscale image as secret data, with $512 \times 512 \times 8 = 2{,}097{,}152$ bits. The tests showed that these bits can be hidden in eight other 512×512 grayscale images, with perfect data extraction and image recovery. All eight images have the same high capacity, with ER $= 512 \times 512 \times 8/512 \times 512 = 8$ bpp. To demonstrate the superiority of our scheme's capacity, we compared it with other state-of-the-art methods, including the quantum RDH scheme [42]. The results, presented in Table I, use the values reported to ensure a fair comparison. Our scheme has a higher ER than existing schemes, offering greater capacity. This is because other methods rely on vacant room, which depends on pixel links and the embedding algorithm. However, our CNOT-based fusion depends solely on the cover image size, avoiding these limitations and providing a high-capacity solution for RDHEI.

2) Complexity Analysis: To assess the complexity of the quantum circuit, the following two key metrics are used: 1) quantum cost: the number of elementary quantum gates used in the quantum circuit, and 2) time complexity: It denotes the depth or the number of time steps in the execution of the quantum circuit (where the execution of all basis operations in one step is taken as 1 unit).

a) Quantum cost: In our case, the part of the circuit that affects the quantum cost is the encoding of the pixel values according to the positions of the pixels, because this part includes several $\mathbb{C}^{2n}$ (NOT gates). These gates are decomposed into $2(2n-1)$ Toffoli gates and a CNOT gate with $2n-1$ ancillary qubit, where $\mathbb{C}^{2n}$ represents the $2n$ control qubits. In total, the decomposition of the $\mathbb{C}^{2n}$ NOT gate into single and two-qubit basis gates will greatly increase the cost.

This increase in cost can be minimized with the help of the Espresso algorithm. The algorithm takes control-qubit information and returns a circuit that performs the same control task with fewer control qubits. For example, the NEQR circuit in Fig. 9(a), which has 14 Toffoli gates, is optimized and reduced to the circuit in Fig. 9(b), which has only 8 Toffoli gates by applying the Espresso algorithm. As shown in Table III, the quantum cost for the baseline NEQR circuit is 234, which is reduced to 154 after applying the Espresso algorithm - a reduction of 34. 19%. To further optimize the quantum cost, the technique demonstrated in Fig. 9(c) has been deployed along with the Espresso algorithm, resulting in the conversion of our circuit shown in Fig. 9(d) with a quantum cost of 91. This represents an additional 40.9% reduction in the Espresso-optimized circuit and a cumulative reduction of 61.11% from the baseline. Similar reduction patterns are observed for the encryption circuit (cumulative reduction of 67.57%) and the decryption circuit (cumulative reduction of 69.63%).

b) Time complexity: As discussed in [14], the time complexity of the NEQR circuit is less than $\mathcal{O}(qn2^{2n})$. Similarly, in [19], it was shown that the time complexity of the encryption procedure is $\mathcal{O}(n)$, which is the same for the decryption procedure. The quantitative analysis of the time complexity reduction for the encryption and decryption processes is presented in Table III. The baseline NEQR circuit has a time complexity of 147, which is reduced to 93 with the Espresso algorithm (36.73% reduction). After including the ancillary qubit

TABLE I
EMBEDDING RATE (ER) IN DIFFERENT METHODS (BPP)

Image	Yin [43]	Yang [44]	Wang [45]	Hua [46]	Ping [47]	Yao [48]	Wang [49]	Gao [50]	Yu [51]	Luo [42]	Proposed
Peppers	2.187	3.110	0.394	1.861	2.494	2.631	1.511	0.973	2.875	2	**8**
Airplane	3.725	-	0.489	3.508	-	3.894	-	0.966	4.869	2	**8**
Baboon	1.066	1.680	0.400	0.565	1.199	1.375	-	0.836	1.544	2	**8**
Boat	2.132	3.320	0.422	-	-	2.476	1.286	0.961	-	2	**8**
House	2.679	-	0.475	-	-	3.302	1.682	0.962	-	2	**8**
Jetplane	3.030	3.780	-	2.589	3.183	3.535	2.023	0.985	3.635	2	**8**
Lake	1.887	2.650	0.454	-	2.085	2.241	1.235	0.976	-	2	**8**
Barbara	1.806	2.460	0.437	-	2.179	2.478	1.166	0.933	-	2	**8**
Average	2.314	2.833	0.439	2.131	2.228	2.742	1.484	0.949	3.231	2	**8**

TABLE II
EXPERIMENTAL RESULTS OF PSNR (DB) AND SSIM ON TEST IMAGES

Image	Indicator	Peppers	Airplane	Baboon	Boat	House	Jetplane	Lake	Barbara
Original & Encrypted	PSNR	8.9430	9.0713	9.6491	9.4072	8.6914	8.0548	8.2286	9.2423
	SSIM	0.0091	0.0102	0.0102	0.0081	0.0104	0.0093	0.0080	0.0087
Original & Masked	PSNR	8.9563	9.0238	9.6469	9.4208	8.6324	7.9650	8.2746	9.2428
	SSIM	0.0097	0.0103	0.0090	0.0112	0.0086	0.0093	0.0093	0.0101
Encrypted & Masked	PSNR	4.8464	4.8514	4.9031	4.9018	4.7894	4.7399	4.7702	4.8780
	SSIM	-0.9755	-0.9753	-0.9755	-0.9756	-0.9755	-0.9753	-0.9753	-0.9755
Original & Direcly Decrypted	PSNR	7.4034	7.6737	9.5699	8.7184	6.7282	5.2754	5.7696	8.2107
	SSIM	-0.1397	0.3141	-0.6515	-0.3460	-0.2710	-0.1049	-0.3015	-0.3077
Original & Recovered	PSNR	$+\infty$	$+\infty$	$+\infty$	$+\infty$	$+\infty$	$+\infty$	$+\infty$	$+\infty$
	SSIM	1	1	1	1	1	1	1	1

TABLE III
INCREMENTAL AND CUMULATIVE REDUCTION ANALYSIS OF QUANTUM COST AND TIME COMPLEXITY FOR THE NEQR, NEQR WITH ENCRYPTION, AND NEQR WITH ENCRYPTION AND DECRYPTION QUANTUM CIRCUITS FOR AN IMAGE REPRESENTED BY (2). THE CALCULATION IS PERFORMED USING THE QISKIT TRANSPILER WITH AN OPTIMIZATION LEVEL OF 3 AND BASIS GATE SET [CX, ID, RZ, SX, X], WHICH IS THE MOST COMMON BASIS GATE SET FOR IBM'S QUANTUM HARDWARE. "BASELINE" REPRESENTS CIRCUITS WITHOUT ANY CIRCUIT OPTIMIZATION. IN "ESPRESSO", THE VALUES ARE CALCULATED WITH THE ESPRESSO ALGORITHM COMPRESSION METHOD. "OUR APPROACH" COMBINES THE ESPRESSO ALGORITHM WITH THE ANCILLARY QUBIT COMPRESSION METHOD SHOWN IN FIG. 9(C). THE UNIT OF QUANTUM COST IS THE NUMBER OF GATES, AND THE TIME COMPLEXITY IS THE TIME STEPS REQUIRED TO EXECUTE ALL PARALLELIZABLE QUANTUM GATES.

Optimization Stage	**Circuit**	**Quantum Cost**	**Reduction from Previous stage (%)**	**Cumulative Reduction (%)**	**Time Complexity**	**Reduction from Previous stage (%)**	**Cumulative Reduction (%)**
Baseline (Normal)	NEQR	234	–	–	147	–	–
	Encryption	666	–	–	513	–	–
	Decryption	1126	–	–	887	–	–
Espresso	NEQR	154	34.19%	34.19%	93	36.73%	36.73%
	Encryption	374	43.85%	43.85%	247	51.85%	51.85%
	Decryption	597	46.98%	46.98%	403	54.57%	54.57%
Our Approach (Espresso + Ancillary Qubit)	NEQR	91	40.9%	61.11%	52	44.09%	64.63%
	Encryption	216	42.25%	67.57%	134	45.75%	73.88%
	Decryption	342	42.71%	69.63%	215	46.65%	75.76%

compression technique shown in Fig. 9(c) along with the Espresso algorithm, the time complexity is further reduced to 52, an additional reduction of 44.09% and a cumulative reduction of 64.63% from the baseline. Similar reductions of 73.88% and 75.76% are observed in the encryption and decryption circuits, respectively.

These results show that combining the Espresso algorithm with the ancillary qubit compression technique significantly reduces the circuit and time complexity.

C. Encryption Security Analysis

The security of the encryption procedure depends on the keyspace, key sensitivity, and its ability to generate different encrypted images for minor changes in the original image.

1) Statistical Analysis: Statistical features of an image, such as histograms and correlation coefficients of adjacent pixels, are essential to evaluate the effectiveness of encryption schemes against attacks. Histogram analysis reveals that, as shown in Fig. 10 for the Cameraman image, the original and fully recovered image exhibit identical distributions with salient statistical features, while the encrypted and masked images display flatly distributed histograms, making it difficult to identify matching features and thus providing strong resistance to histogram analysis attacks. For correlation analysis, the coefficient C_{xy} measures the relationship between adjacent pixels and is calculated as

$$C_{xy} = \frac{\sum(x_i - x_m)(y_i - y_m)}{\sqrt{\sum(x_i - x_m)^2 \sum(y_i - y_m)^2}},$$

where x_i and y_i are gray pixel intensities of adjacent pairs, and $x_m = \frac{1}{N}\sum x_i$ and $y_m = \frac{1}{N}\sum x_i$ denote the mean pixel intensities. The values of C_{xy} near 1 indicate a strong correlation between adjacent pixels, whereas values near 0 indicate minimal correlation and effective encryption. Table IV lists the correlation coefficients for eight original, encrypted, and marked

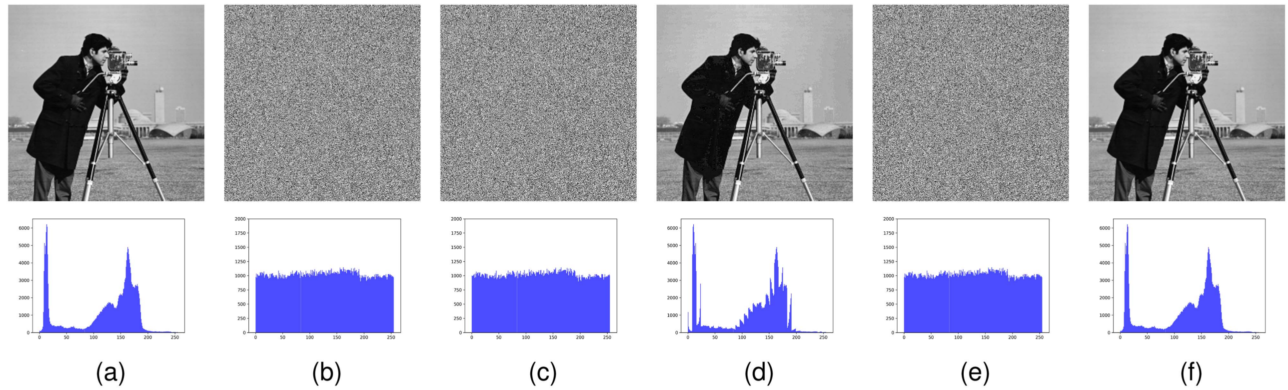


Fig. 10. Different stages of our proposed scheme along with histogram analysis: (a) Original cameraman image, (b) Encrypted cameraman image using GAT and LM. The parameters used for encryption are $P = 57$, $Q = 74$, $s = 1$, $t = 1$ in the GAT and $L_0 = 0.55579243169496\,03$, $\delta = 3.98161887277912\,15$ in the LM. (c) Data-embedded encrypted image, (d) Directly decrypted image, (e) Data-extracted encrypted image, and (f) Fully recovered original image.

TABLE IV
CORRELATION ANALYSIS OF ORIGINAL, ENCRYPTED, AND MASKED IMAGES IN DIFFERENT DIRECTIONS

Image	State	Horizontal	Vertical	Diagonal
Pappers	Original	0.9767	0.9792	0.9639
	Encrypted	0.0010	0.0016	0.0002
	Masked	0.0010	0.0016	0.0002
Airplane	Original	0.9463	0.9458	0.8961
	Encrypted	0.0007	0.0016	-0.0008
	Masked	0.0007	0.0016	-0.0008
Baboon	Original	0.8665	0.7587	0.7261
	Encrypted	-0.0002	0.0003	0.0028
	Masked	-0.0002	0.0003	0.0028
Boat	Original	0.9381	0.9713	0.9221
	Encrypted	0.0007	0.0008	0.0015
	Masked	0.0007	0.0008	0.0015
House	Original	0.9484	0.9577	0.9135
	Encrypted	0.0007	0.0043	0.0012
	Masked	0.0007	0.0043	0.0012
Jetplane	Original	0.9663	0.9641	0.9370
	Encrypted	-0.0007	0.0053	0.0010
	Masked	-0.0007	0.0053	0.0010
Lake	Original	0.9750	0.9715	0.9577
	Encrypted	-0.0019	0.0009	0.0026
	Masked	-0.0019	0.0009	0.0026
Barbara	Original	0.8597	0.9590	0.8418
	Encrypted	0.0023	0.0033	-0.0004
	Masked	0.0023	0.0033	-0.0004

images in horizontal, vertical, and diagonal orientations. The results show that the correlations in the encrypted and masked images are significantly weaker than those in the original images. These results demonstrate that the proposed scheme obscures statistical features, preventing attackers from obtaining useful information through correlation analysis.

2) Information Entropy Analysis: Information entropy quantifies uncertainty or randomness in an image, indicating the amount of information it carries. For 256 grayscale level images, encrypted, masked, or cipher images are expected to approach the ideal value of 8, indicating high randomness that enhances security against eavesdroppers extracting useful information. The entropy is calculated as

$$I_E = -\sum_{t=0}^{255} P(t) \log_2 P(t), \tag{10}$$

Where $P(t)$ denotes the probability of the occurrence of a grayscale value t. Table V presents the calculated entropy values for our various images shown in Fig. 11. The results in Table V show that encrypted and marked images exhibit higher entropy values than the originals, approaching 8. This indicates that the proposed algorithm exhibits high information entropy, making it difficult for an eavesdropper to extract useful information.

TABLE V
INFORMATION ENTROPY OF THE ORIGINAL, ENCRYPTED, AND MASKED IMAGES

Image	Original Image	Encrypted Image	Masked Image
Peppers	7.5936	7.9984	7.9984
Airplane	4.0044	7.9964	7.9964
Baboon	7.3583	7.9978	7.9978
Boat	7.1913	7.9977	7.9977
House	7.2334	7.9989	7.9989
JetPlane	6.7024	7.9982	7.9982
Lake	7.4842	7.9991	7.9991
Barbara	7.4664	7.9983	7.9983

3) Brute-Force Attack:: The keyspace is the set of all possible key combinations in an encryption system, where one key can perfectly decrypt the system. The keyspace increases with the number of bits (n), and can be represented as:

$$keyspace(KS) \propto 2^n.$$

For GAT, n_P, n_Q, n_s, n_t-qubit have been used to store the parameters P, Q, s, and t, respectively. Hence, the keyspace for GAT is $KS_A = 2^{n_P+n_Q+n_s+n_t}$. For LM, n_{L_0} and n_δ-qubit for L_0 and δ, have been used. Hence, the keyspace is $KS_L = 2^{n_{L_0}+n_\delta}$. The total keyspace is $KS_T = KS_A + KS_L$.

To secure the image, the encryption parameters $(P$, Q, s, t, L_0, $\delta)$ should be chosen so that a brute-force attack (attempting all possible key combinations) cannot break encryption in a

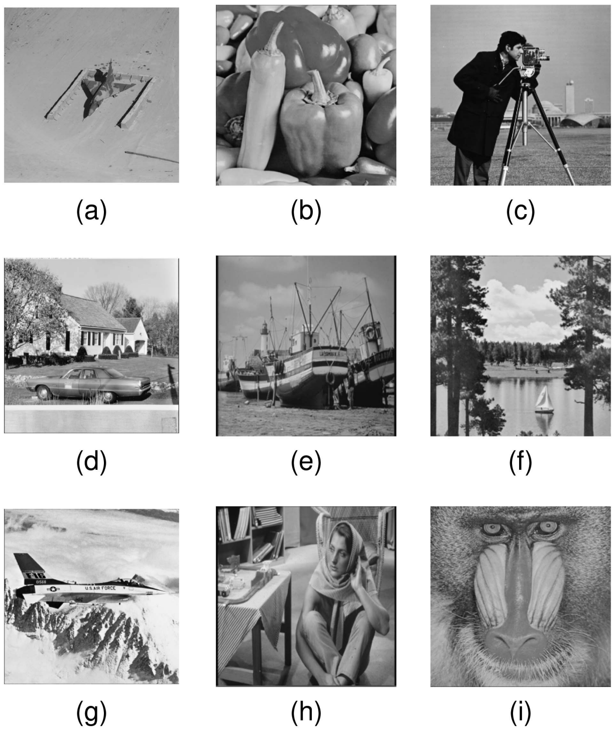


Fig. 11. Test images: (a) Jetplane (b) Peppers (c) Cameraman (d) Crowd (e) Boat (f) Lake (g) Airplan (h) Barbara (i) Baboon.

feasible amount of time. L_0 and δ are infinite decimal values that make the keyspace effectively infinite [19]. Therefore, the encrypted image is highly resistant to brute-force attacks.

4) Differential Attack: Comparison analysis of differential attacks is a key method of evaluating security. It examines how small changes in the original key affect encrypted images. A secure encryption scheme must exhibit high key sensitivity, meaning that even minor alterations to the encryption key produce completely different ciphertexts. We measure this using two standard metrics: NPCR (number of pixels change rate) and UACI (unified average change intensity), defined as:

$$\text{NPCR} = \frac{\sum_{i=1}^{Z_1}\sum_{j=1}^{Z_2} D(i,j)}{Z_1 \times Z_2} \times 100\% \tag{11}$$

$$\text{UACI} = \frac{\sum_{i=1}^{Z_1}\sum_{j=1}^{Z_2}(|x_{i,j} - x'_{i,j}|/255)}{Z_1 \times Z_2} \times 100\% \tag{12}$$

Where $Z_1 \times Z_2$ represents the image dimensions, $x_{i,j}$ and $x'_{i,j}$ denote pixel values at position (i,j) in two ciphertexts encrypted with different keys, and $D(i,j)$ is a bipolar function equal to 1 if $x_{i,j} \neq x'_{i,j}$ and 0 otherwise. For secure encryption, the NPCR and UACI values should approach their theoretical limits of 99.61% and 33.46%, respectively [44]. To assess the sensitivity of the keys, we slightly varied the original encryption key $L_0 = 0.5557924316949603$. We created 11 modified keys with different offsets of $\pm n \times 10^{-16}$ (where $n \in \{1, 10, 20, 30, 40, 50\}$). Both the original key and the modified keys are used to encrypt each test image, and the NPCR and UACI metrics are calculated by comparing ciphertext pairs. The information is presented in Table VI, showing that the mean NPCRs and UACIs across all test images are tidy and close to ideal values, with minimal variation between the minimum and maximum recorded values. These results indicate that the proposed encryption scheme ensures a high level of key sensitivity and resistance to differential attacks.

TABLE VI
RESULTS OF NPCR AND UACI FOR DIFFERENTIAL ATTACK ANALYSIS ON TEST IMAGES (%)

Image	Minimum	Maximum	Mean
	NPCR/UACI	NPCR/UACI	NPCR/UACI
Peppers	99.4507/32.7203	99.7070/33.2145	99.6338/33.0265
Baboon	99.4629/33.4886	99.5979/33.3714	99.6122/33.4144
Airplane	99.5249/33.5522	99.7675/33.1673	99.6930/33.3800
Boat	99.5916/33.6268	99.6947/33.3293	99.6728/33.4830
House	99.5364/32.7532	99.6801/33.0700	99.6290/32.9028
Jetplane	99.6070/33.1646	99.7370/33.3832	99.6857/33.2080
Lake	99.4630/33.6347	99.7053/33.2099	99.6535/33.3467
Barbara	99.5490/32.7745	99.6542/33.2395	99.6293/33.0110

TABLE VII
COMPARISON OF SECURITY EVALUATION OF NEQRX VS. OTHER METHODS

Methods	Correlation coefficient			Entropy	PSNR	SSIM
	H	V	D			
Yu [52]	0.0074	0.0006	0.0020	7.9985	8.7567	0.0376
Yang [44]	0.0028	0.0021	-	7.9982	8.6214	-
Fu [53]	0.0010	0.0035	0.0027	7.9993	8.5398	-
Ping [47]	0.0071	0.0078	0.0109	7.9835	-	-
Gao [50]	0.0046	0.0083	-	7.9906	8.0126	0.0088
Kumar [54]	0.0251	0.0199	0.0241	7.9970	9.1902	0.0108
Proposed	0.0023	0.0032	0.0005	7.9983	9.3449	0.0106

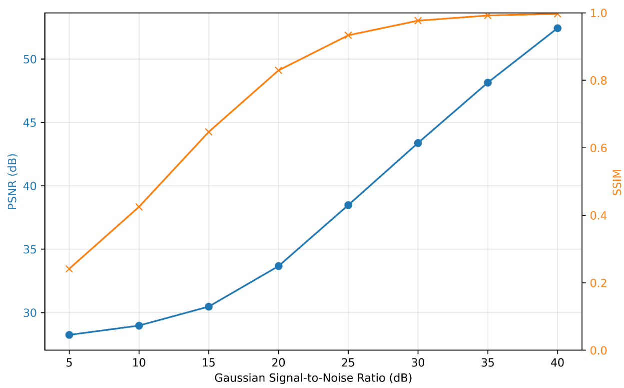
5) Comparison With Other Methods: Here, we compare our method with other security verification methods. Table VII presents the correlation coefficient and the information entropy for the masked images generated by our encryption scheme, together with the PSNR and SSIM values for the original images and their masked versions. In fairness, the information in the table is taken directly from documented works or averaged data. In particular, we demonstrated the correlation value of our scheme as the absolute value. For correlation values, our scheme is closer to 0 than others, indicating a better ability to avoid correlation attacks. The information entropy values are lower than those reported in [52] and [53], but around 8, and higher than those reported in [44], [47], [50], and [54], indicating a good entropic hiding of the image content. The PSNR and SSIM values are low (under 10 dB and close to 0), indicating a low similarity between the original and masked images.

D. Image Size Scalability

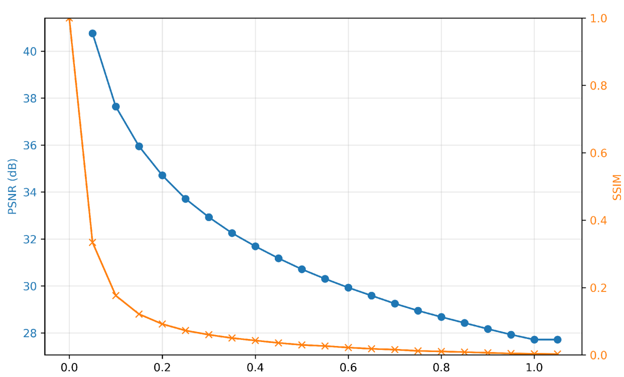
We tested six image sizes, ranging from 8 $\times$ 8 to 256 $\times$ 256 pixels. The results shown in Table VIII show that the entropy increases from 5.7812 (8 $\times$ 8) to 7.9948 bits/pixel (256 $\times$ 256), approaching the theoretical maximum of 8 bits/pixel for larger images. The correlation coefficients remain near zero across all image sizes (from -0.0808 to 0.0075), confirming that the decorrelation process is not scale-dependent. The PSNR values range from 9.88 to 11.16 dB, and the SSIM values remain low, indicating substantial visual dissimilarity between the original and encrypted images. The embed capacity increases linearly from 512 bits to 524,288 bits, indicating that

TABLE VIII
SCALABILITY ANALYSIS: QUALITY AND SECURITY METRICS ACROSS MULTIPLE IMAGE SIZES

Image Size	Correlation coefficient			Entropy	PSNR	SSIM	Embedding
	H	V	D				Capacity
8×8	-0.0441	-0.0731	-0.0063	5.7812	11.1630	-0.1359	512
16×16	-0.0291	-0.0808	0.0042	7.1774	10.1359	0.0429	2048
32×32	-0.0602	0.0075	-0.0674	7.7674	10.7295	0.0076	8192
64×64	0.0006	-0.0000948	-0.0046	7.9569	10.1378	0.0148	32768
128×128	-0.0019	-0.0043	0.0050	7.9856	9.9901	0.0114	131072
256×256	0.0042	0.0037	0.0055	7.9948	9.8794	0.0094	524288



(a)



(b)

Fig. 12. PSNR and SSIM of decrypted images after adding (a) Gaussian noise at various SNRs and (b) salt-and-pepper noise at various densities. Error bars show standard deviation over the test set.

the algorithm can encode more data in larger images. This indicates that encryption security increases with larger images due to higher entropy, while the correlation remains low—close to zero—supporting continued evidence of improved encryption efficiency.

E. Noise Analysis

1) Robustness to Classical Noise: To evaluate the resilience of the proposed method against common image degradations, Gaussian noise (varying signal-to-noise ratio, SNR) and salt-and-pepper noise (varying density) are added to encrypted images prior to decryption. Figs. 12(a) and (b) plot PSNR and SSIM of the decrypted images as functions of SNR and density, respectively.

For Gaussian noise, SNR values from 5 to 40 dB were tested. PSNR increases from 28.23 dB at 5 dB to 52.44 dB at 40 dB, while SSIM increases from 0.24 to 0.997. Even at moderate noise levels (SNR = 15 dB), PSNR already exceeds 30 dB, and SSIM exceeds 0.65, indicating excellent visual detail recovery.

For salt-and-pepper noise, densities from 0.00 to 1.05 were evaluated. At density=0, PSNR is infinite (perfect reconstruction) and SSIM = 1. As density increases to 1.00, PSNR gradually falls to 27.71 dB and SSIM to 0.0033, yet the decrypted images still retain recognizable structure even at extreme contamination.

These results confirm that the encryption method maintains high image fidelity under a wide range of classical noise attacks, demonstrating strong antinoise robustness.

2) Quantum Hardware Noise: We examine six different types of noisy environments: amplitude-damping, phase-damping, bit-flip, phase-flip, bit-phase-flip, and depolarizing. Kraus operators characterize these noisy channels [55], and a detailed explanation of these operators is provided in the supplementary file. The effect of this noise on the real quantum state $|\psi\rangle$ is shown in Fig. 13 and can be quantified by calculating the fidelity, which is given by

$$F(\rho, \epsilon(\rho)) = \left[Tr\sqrt{\sqrt{\rho}\epsilon(\rho)\sqrt{\rho}} \right]^2, \qquad (13)$$

where $\rho = |\psi\rangle \langle\psi|$ is the pure state, $\epsilon(\rho)$ is the noisy state and $F(\rho, \epsilon(\rho)) \in [0, 1]$, where 1 means that both states are identical and 0 means that the states are entirely different.

The optimized circuit reflects a depth–qubit trade-off. Reducing the gate count and circuit depth improves fidelity on noisy hardware by limiting the accumulated probability of gate error and decoherence, which is significant because multi-control NEQR gates expand into many one- and two-qubit basis gates after transpilation (Section V-B2). The ancillary qubit introduces a small overhead, extra decoherence exposure and possible routing cost on hardware with limited connectivity, but the reductions in Table III (up to 69.63% in quantum cost and 75.76% in time complexity for the decryption circuit) far exceed this one-qubit cost. Hence, for near-term hardware dominated by gate-accumulation and circuit-duration errors, the design favors a lower-depth implementation at the cost of a single ancillary qubit and is expected to improve practical fidelity.

TABLE IX
PERFORMANCE BENCHMARKING OF NEQRX VS. [19]

Comparison Metric	[19]	NEQRX (Our Approach)
Circuit Optimization	Not addressed	Up to 69.63% quantum cost reduction
Data Hiding Capability	Encryption only	Integrated RDH with 35 dB PSNR at 3 bpp
Scalability	Limited by circuit complexity	Enhanced through multi-stage optimization
Practical Implementation	High resource requirements	Minimized resource usage for NISQ-era hardware
Time Complexity	Standard $O(qn2^{2n})$	Up to 75.76% reduction from baseline

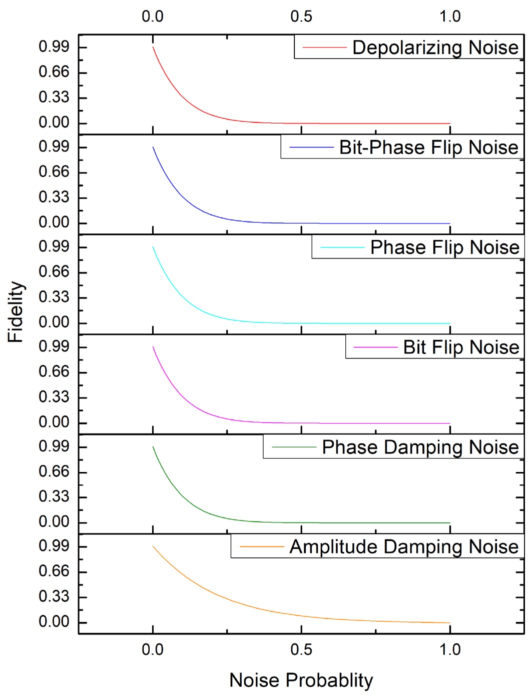


Fig. 13. Noisy analysis of the quantum state shown in (2) for six different types of noisy channels. Here, the fidelity variation comes out to be similar for all the noisy environments except the amplitude-damping noisy environment.

F. Comparative Analysis With Existing GAT+LM Quantum Image Encryption Schemes

Although GAT and LM were alternative foundational approaches to quantum image encryption, our NEQRX offers several notable innovations that lead to key improvements over [19]. First, our two-stage approach reduces the complexity of the quantum circuit. When combined, our approach achieves cumulative reductions of up to 69.63% and 75.76% in time complexity across the entire encryption-decryption quantum circuit. For 2 x 2 NEQR circuits, the total gate count decreased from 234 to 91, a reduction of 61.11.

Importantly, these optimizations and functional enhancements do not compromise security. NEQRX's security—key sensitivity, diffusion, and permutation—is inherited from the GAT+LM chaotic construction (Section V-C), while the quantum contribution is architectural: a reversible, low-cost realization over NEQR image states together with integrated RDH. This differs from natively quantum primitives based on entanglement, measurement disturbance, or quantum key distribution, which are intended for key establishment. NEQRX instead targets an encrypted quantum image representation and reversible data embedding, preserving the security behavior of the classical cipher while enabling quantum-compatible, low-depth encryption. The fidelity analysis across six quantum noise models further demonstrates NEQRX's noise resilience. Taken together, these enhancements make NEQRX a more efficient, versatile, and practical quantum image processing system than the existing scheme of [19].

VI. CONCLUSION

This paper introduces an efficient implementation scheme for a Novel Enhanced Quantum Representation (NEQR) of images, along with a comprehensive quantum image encryption and decryption procedure. The proposed algorithms demonstrate superior performance in digital image representation and robust encryption due to their high measurement accuracy, chaotic outputs, large keyspaces, and sensitivity to small changes. To address the increasing circuit complexity arising from decomposing multi-control-NOT gates into one- and two-qubit operations, we introduce a two-stage optimization framework. This approach achieves comparative reductions of up to 42.71% in quantum cost and 46.65% in time complexity for the complete decryption circuit of a 2×2 test image compared to the Espresso algorithm, which requires only one additional qubit, a favorable trade-off between quantum storage requirements and computational efficiency.

Furthermore, we developed a Reversible Data Hiding (RDH) algorithm. We compared it with other state-of-the-art RDHEI methods, focusing on PSNR and embedding capacity (bpp). Although PSNR was initially lower at low embedding rates, it stabilized and outperformed all competing methods at higher embedding rates, demonstrating superior embedding capacity and maintaining high image quality.

A key challenge for the practical validation of this and other quantum image processing schemes is the extensive simulation time required for large-scale images: although quantum simulators can process high-resolution images such as 1024×1024, the execution time per image exceeds 40 hours due to the exponential scaling of quantum circuit simulation, which restricts comprehensive validation to smaller image dimensions. This bottleneck underscores the importance of the complexity-reduction techniques introduced in this work. Future work will focus on the following directions: First, validation of the framework on backend-aware execution under realistic hardware connectivity, device calibration, and measurement-error mitigation. Second, generalization of the ancillary-qubit compression to larger NEQR images and other quantum image representations, thereby characterizing when the depth reduction outweighs the added qubit cost. Finally, extending the security analysis to broader threat models involving imperfect quantum hardware, noisy measurements, and partial key access. Together, these steps would bridge the present simulator-based framework toward deployment on future high-fidelity or fault-tolerant quantum processors.

In conclusion, our research demonstrated that NEQR, combined with the proposed quantum image encryption method and the novel RDH algorithm, form an efficient, secure, and accurately measurable quantum image processing system. The combined approaches ensure high data embedding capacity, reduced circuit complexity, and robust resistance to attacks.

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